1D CNN — Architecture Variant Comparison

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Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	325 (train pool 292, hold-out 33)
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	10 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	7 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

Sensor grid (✓ enabled, ✗ disabled, — not present)

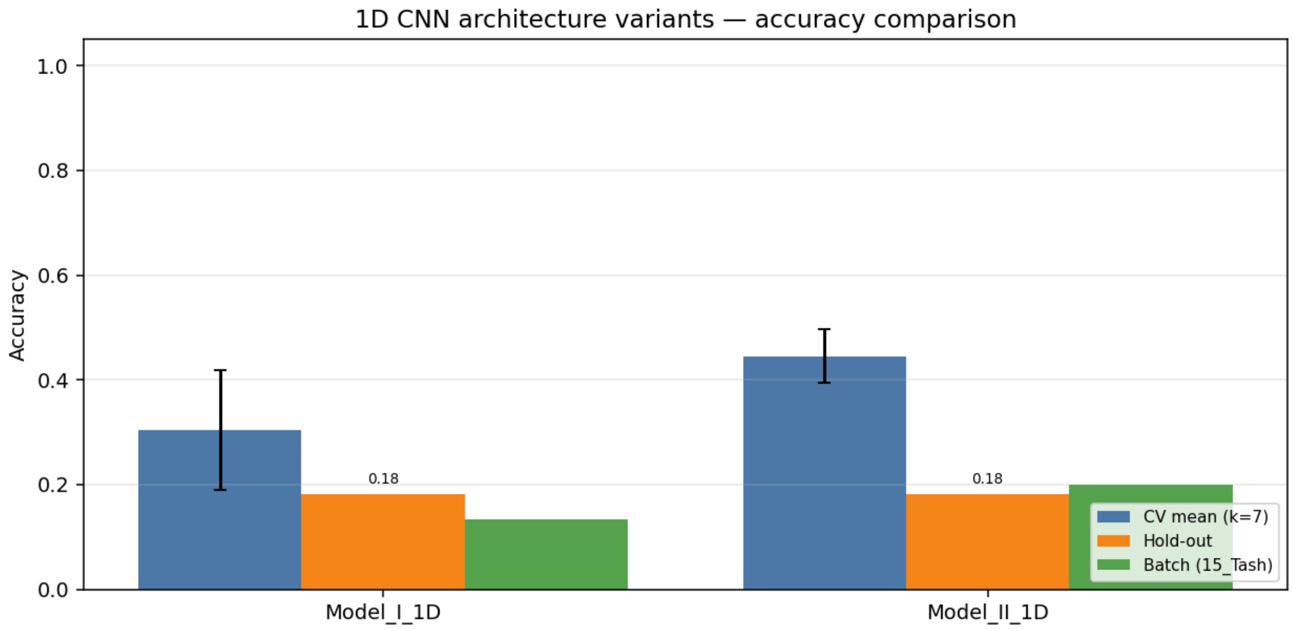
Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✗	✓	✗	✓
left index	✗	✓	✗	✓
left middle	✗	✓	✗	✓
left ring	✗	✓	✗	✓
left pinky	✗	✓	✗	✓
left wrist	✗	—	—	—
right palm	✓	—	—	—
right thumb	✗	✓	✗	✓
right index	✗	✓	✗	✓
right middle	✗	✓	✗	✓
right ring	✗	✓	✗	✓
right pinky	✗	✓	✗	✓
right wrist	✗	—	—	—

Enabled: 22 · Disabled: 22 · Modalities: Flex → 10 channels total.

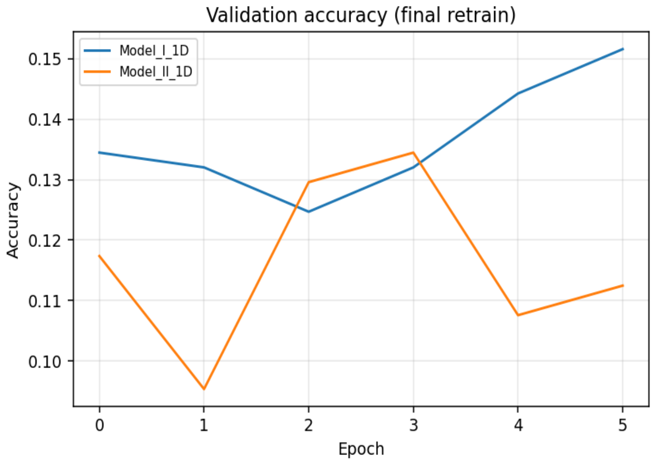
Variant comparison (summary)

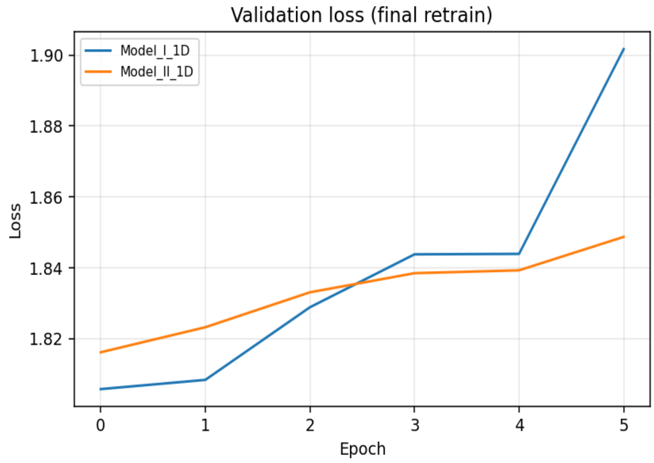
Variant	Params	CV mean (k=7)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Model_I_1D	117,030	0.3046	0.1146	0.1818	1.7968	0.1333	4/30
Model_II_1D	72,326	0.4451	0.0512	0.1818	1.7873	0.2000	6/30

Best on hold-out: **Model_I_1D** · Best on CV: **Model_II_1D**



Training curves (final retrain on full training pool)





Variant: Model_I_1D

1-D adaptation of published Model I: Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

Trainable params	117,030
CV mean ± std	0.3046 ± 0.1146
Hold-out test acc / loss	0.1818 · 1.7968
Batch-test acc	0.1333 (4/30)

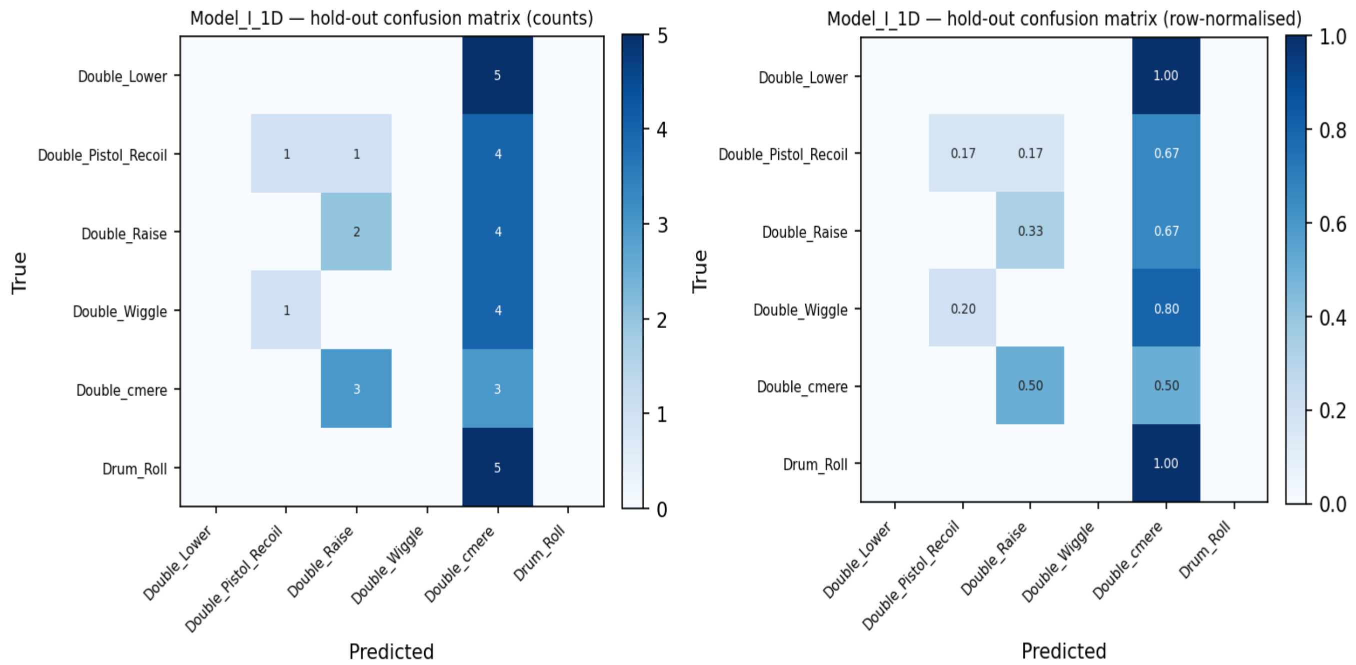
Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	0.1905	0.2857	0.1905
2	1750	42	0.4524	0.5000	0.4524
3	1750	42	0.5000	0.5000	0.3571
4	1750	42	0.2381	0.3571	0.3571
5	1750	42	0.1905	0.2857	0.2857
6	1757	41	0.2683	0.2927	0.1951
7	1757	41	0.2927	0.3659	0.3659

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.000	0.000	0.000	5
Double_Pistol_Recoil	0.500	0.167	0.250	6
Double_Raise	0.333	0.333	0.333	6
Double_Wiggle	0.000	0.000	0.000	5
Double_cmere	0.120	0.500	0.194	6
Drum_Roll	0.000	0.000	0.000	5
macro avg	*0.159*	*0.167*	*0.129*	*33*
weighted avg	*0.173*	*0.182*	*0.141*	*33*

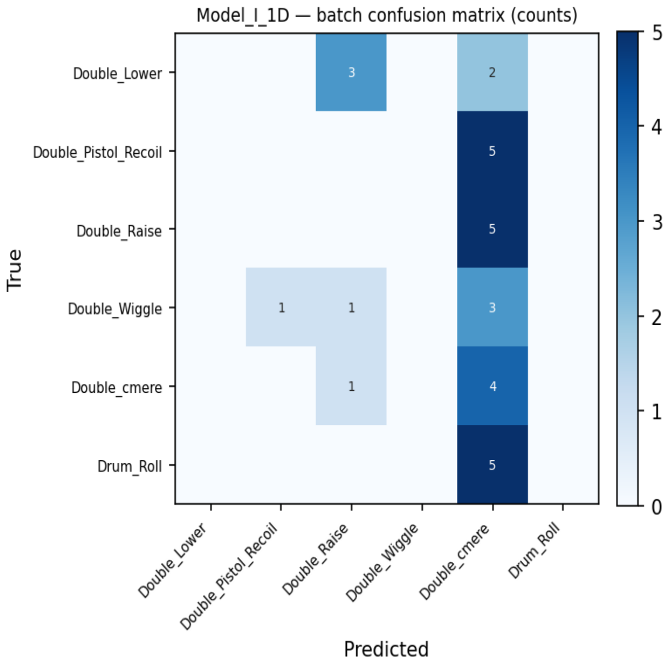
Hold-out confusion matrix

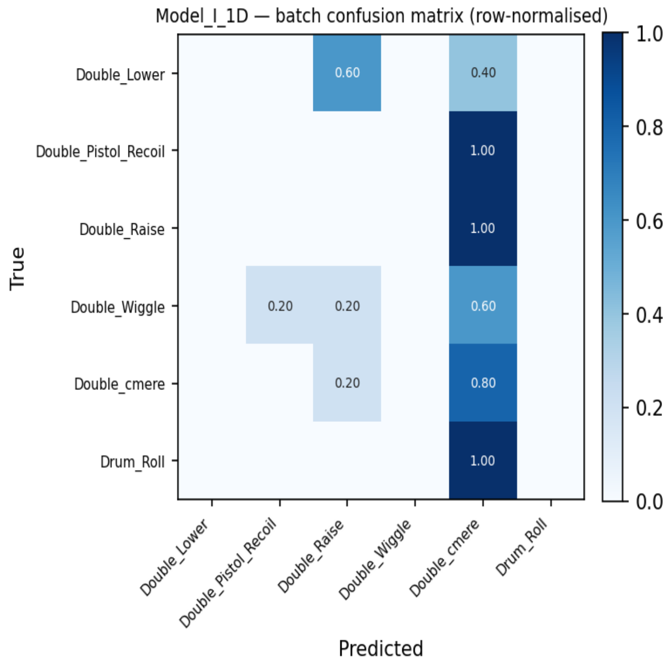


Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Pistol_Recoil	0	5	0.000
Double_Raise	0	5	0.000
Double_Wiggle	0	5	0.000
Double_cmere	4	5	0.800
Drum_Roll	0	5	0.000
Overall	**4**	**30**	**0.133**

Batch confusion matrix (15_Tash)


Model_I_1D — batch confusion matrix (counts)
True
Predicted
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
3
2
5
5
1
1
3
1
4
5
0
1
2
3
4
5


Model_I_1D — batch confusion matrix (row-normalised)
True
Predicted
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
0.60
0.40
1.00
1.00
0.20
0.20
0.60
0.20
0.80
1.00
0.0
0.2
0.4
0.6
0.8
1.0

Variant: Model_II_1D

1-D adaptation of published Model II (CNN+LSTM): Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

Trainable params	72,326
CV mean ± std	0.4451 ± 0.0512
Hold-out test acc / loss	0.1818 · 1.7873
Batch-test acc	0.2000 (6/30)

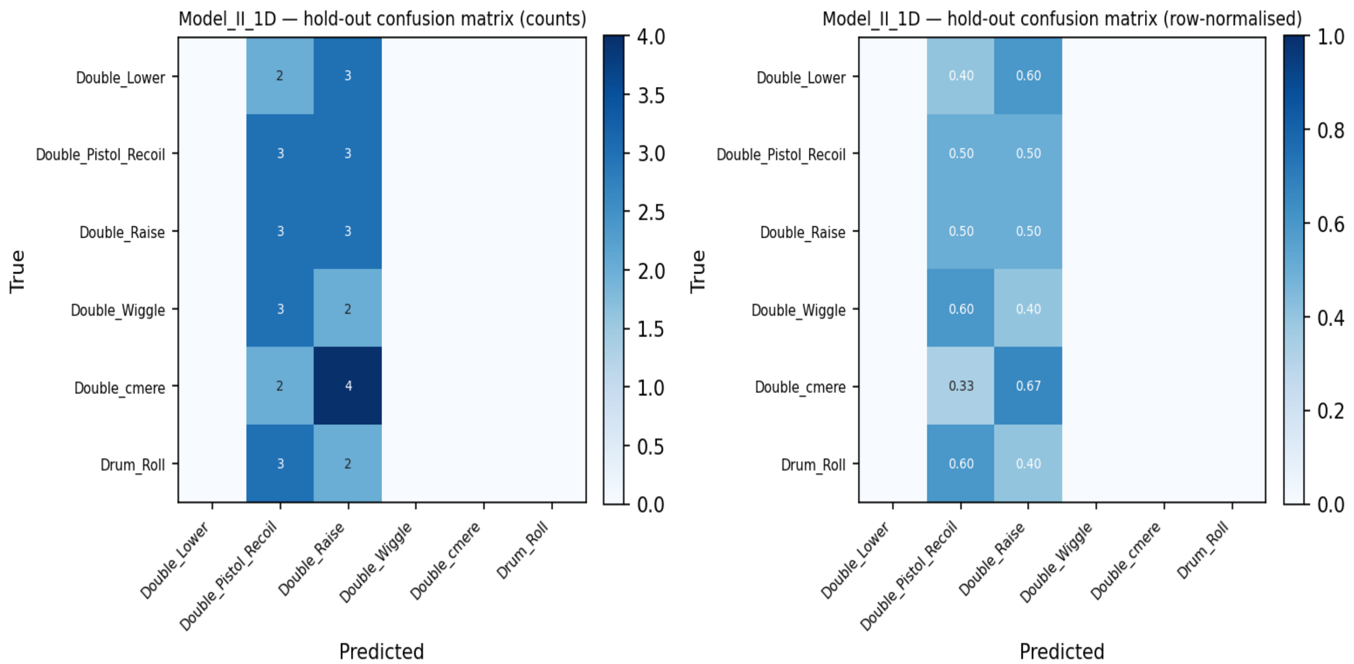
Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	0.4524	0.5000	0.4524
2	1750	42	0.3810	0.5238	0.4524
3	1750	42	0.5476	0.5714	0.5000
4	1750	42	0.4524	0.5714	0.4762
5	1750	42	0.4286	0.5238	0.5238
6	1757	41	0.4634	0.4878	0.4390
7	1757	41	0.3902	0.4146	0.3902

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.000	0.000	0.000	5
Double_Pistol_Recoil	0.188	0.500	0.273	6
Double_Raise	0.176	0.500	0.261	6
Double_Wiggle	0.000	0.000	0.000	5
Double_cmere	0.000	0.000	0.000	6
Drum_Roll	0.000	0.000	0.000	5
macro avg	*0.061*	*0.167*	*0.089*	*33*
weighted avg	*0.066*	*0.182*	*0.097*	*33*

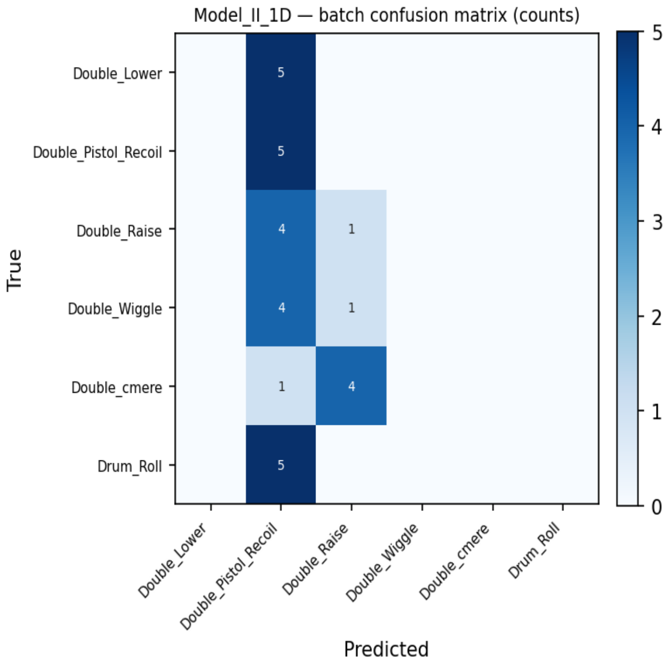
Hold-out confusion matrix

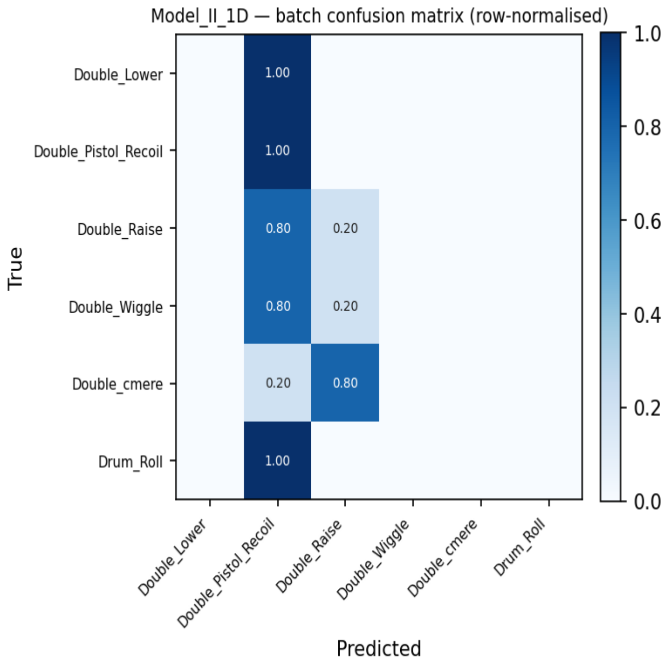


Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	1	5	0.200
Double_Wiggle	0	5	0.000
Double_cmere	0	5	0.000
Drum_Roll	0	5	0.000
Overall	**6**	**30**	**0.200**

Batch confusion matrix (15_Tash)


Model_II_1D — batch confusion matrix (counts)
True
Predicted
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
5
5
4
1
4
1
1
4
5
0
1
2
3
4
5


Model_II_1D — batch confusion matrix (row-normalised)
True
Predicted
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
1.00
1.00
0.80
0.20
0.80
0.20
0.20
0.80
1.00
0.0
0.2
0.4
0.6
0.8
1.0